LAPORAN PRAKTIKUM
WORKSHOP SISTEM INFORMASI WEB SERVER
(TIF330805)
SEMESTER III



Acara 18
JSON encoding dan HTTP status code

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A. Tujuan Pengujian

Pengujian dilakukan untuk memastikan endpoint API mahasiswa dapat menjalankan operasi GET, POST, PUT, dan DELETE dengan benar serta menerapkan autentikasi JSON Web Token (JWT) pada endpoint yang memerlukan akses terproteksi.

Pengujian dilakukan menggunakan **curl.exe** melalui terminal PowerShell sebagai alternatif dari Postman.

1. Metode GET

```
$base = "http://localhost/BKPM/Acara17/si-
akademik/api/mahasiswa"
curl.exe -i $base
```

2. Metode POST

```
$body = '{"nim":"2024004","nama":"Budi
Uji","email":"budiuji@mail.com","jurusan":"Teknik
Informatika"}'
[System.IO.File]::WriteAllText(
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X POST $base `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

3. Metode PUT

```
$body = '{"nim":"2024001","nama":"Budi Santoso
Updated","email":"budi@mail.com","jurusan":"Sistem
Informasi"}'
[System.IO.File]::WriteAllText(
```

```
    "$PWD\body.json",
    $body,
    [System.Text.UTF8Encoding]::new($false)
)
curl.exe -i -X PUT "${base}?id=1" `
  -H "Content-Type: application/json" `
  --data-binary "@body.json"
```

4. Metode DELETE

```
curl.exe -i -X DELETE "${base}?id=1"
```

- **$base** digunakan untuk menyimpan URL endpoint API mahasiswa agar dapat digunakan kembali pada perintah berikutnya.
- **http://localhost** menunjukkan bahwa aplikasi dijalankan pada komputer lokal.
- **/BKPM/Acara17/si-akademik/api/mahasiswa** merupakan alamat endpoint yang menangani data mahasiswa
- **curl.exe** digunakan untuk mengirim permintaan HTTP ke server.
- **-X POST** menentukan bahwa permintaan HTTP menggunakan metode POST untuk mengirim data baru ke API.
- **-X PUT** menentukan bahwa permintaan HTTP digunakan untuk memperbarui data.
- **-X DELETE** menentukan bahwa permintaan HTTP digunakan untuk menghapus data.
- **-i** digunakan untuk menampilkan header respons HTTP beserta isi respons dari server.
- **-H "Content-Type: application/json**" memberi tahu server bahwa isi permintaan menggunakan format JSON.
- **--data-binary "@body.json"** mengirim isi file body.json sebagai badan permintaan (request body).

B. Hasil Pengujian

```
PS C:\laragon\www\BKPM> cd Acara18
PS C:\laragon\www\BKPM\Acara18> $base = "http://localhost/BKPM/Acara18/si-akademik/public/mahasiswa"
PS C:\laragon\www\BKPM\Acara18> curl.exe -s "$base" | ConvertFrom-Json | ConvertTo-Json -Depth 10
{
    "success":  true,
    "message":  "Data berhasil diambil.",
    "data":  [
                 {
                     "id":  3,
                     "nim":  "2201001",
                     "nama":  "Andi Saputra",
                     "prodi":  "Teknik Informatika",
                     "created_at":  "2026-10-21 00:18:40"
                 },
                 {
                     "id":  2,
                     "nim":  "2024002",
                     "nama":  "Siti Aminah",
                     "prodi":  "Sistem Informasi",
                     "created_at":  "2026-10-20 23:15:14"
                 }
             ],
    "timestamp":  "2026-10-21T02:10:19+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

1. Pengujian menggunakan method GET

```
PS C:\laragon\www\BKPM\Acara18> $body = '{"nim":"220100101","nama":"Andi Saputra","prodi":"Teknik Informatika"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X POST "$base" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 201 Created
Date: Wed, 21 Oct 2026 02:14:54 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 153
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil ditambahkan.",
    "data": {
        "id": "4"
    },
    "timestamp": "2026-10-21T02:14:54+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

2. Pengujian menggunakan method POST

```
PS C:\laragon\www\BKPM\Acara18> $body = '{"nim":"220100102","nama":"Andi Saputra Update","prodi":"Sistem Informasi"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X PUT "${base}?id=4" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 02:16:27 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 150
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil diperbarui.",
    "data": {
        "id": 4
    },
    "timestamp": "2026-10-21T02:16:27+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

3. Pengujian menggunakan method PUT

```
PS C:\laragon\www\BKPM\Acara18> curl.exe -i -X DELETE "${base}?id=4"
HTTP/1.1 200 OK
Date: Wed, 21 Oct 2026 02:17:13 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 147
Content-Type: application/json; charset=utf-8

{
    "success": true,
    "message": "Data berhasil dihapus.",
    "data": {
        "id": 4
    },
    "timestamp": "2026-10-21T02:17:13+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

4. Pengujian menggunakan method DELETE

```
PS C:\laragon\www\BKPM\Acara18> curl.exe -i "${base}?id=4"
HTTP/1.1 404 Not Found
Date: Wed, 21 Oct 2026 02:17:29 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 138
Content-Type: application/json; charset=utf-8

{
    "success": false,
    "message": "Data mahasiswa tidak ditemukan.",
    "data": null,
    "timestamp": "2026-10-21T02:17:29+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

5. Pengujian Skenario gagal id tidak ditemukan

```
PS C:\laragon\www\BKPM\Acara18> $body = '{}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X POST "$base" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 400 Bad Request
Date: Wed, 21 Oct 2026 02:18:02 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 146
Connection: close
Content-Type: application/json; charset=utf-8

{
    "success": false,
    "message": "Field nim, nama, dan prodi wajib diisi.",
    "data": null,
    "timestamp": "2026-10-21T02:18:02+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

6. Pengujian Skenario gagal Data kosong

```
PS C:\laragon\www\BKPM\Acara18> $body = '{"nim":"12345","nama":"Uji NIM Salah","prodi":"Teknik Informatika"}'
>>
>> [System.IO.File]::WriteAllText(
>>     "$PWD\body.json",
>>     $body,
>>     [System.Text.UTF8Encoding]::new($false)
>> )
>>
>> curl.exe -i -X POST "$base" `
>>   -H "Content-Type: application/json" `
>>   --data-binary "@body.json"
HTTP/1.1 422 Unprocessable Entity
Date: Wed, 21 Oct 2026 02:21:18 GMT
Server: Apache/2.4.54 (Win64) OpenSSL/1.1.1q PHP/8.1.10
X-Powered-By: PHP/8.1.10
Content-Length: 145
Content-Type: application/json; charset=utf-8

{
    "success": false,
    "message": "Format NIM tidak valid, harus 9 digit.",
    "data": null,
    "timestamp": "2026-10-21T02:21:18+00:00"
}
PS C:\laragon\www\BKPM\Acara18>
```

7. Pengujian skenario gagal Format NIM salah